

Application Architectures: (i) Presentation / user interface (ii) business-logic (iii) data access.

SQL Injection: SQL injection is a security vulnerability that occurs when untrusted data is improperly inserted into a SQL query, allowing an attacker to manipulate the query logic or gain unauthorized access to the database.

(1) Exploiting SQL injection: Attackers typically exploit SQL injection by inserting malicious SQL code into input fields or parameters into a SQL query. Ex: login form where the username and password are concatenated into an SQL query without proper sanitization.

`SELECT * FROM users WHERE username = 'input_username' AND password = 'input_password';` An attacker can pass ' OR 1=1--' as username and no password. Then,

`[SELECT * FROM users WHERE username = "OR 1=1--" AND password = "];` The OR 1=1 condition always true